Text analysis for data science · Research poster

Trump, Iran & the language of conflict

Six rhetorical frames in Truth Social posts before, during, and after selected conflict dates.

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Repository snapshot
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87 unique posts scored · **26** selected conflict start dates · **129** post / event-date pairs · **6** rhetorical frames

Principal finding

Victory scores peak on event days. Threat scores decline across windows.

Across all available date groups, mean Victory scores rise from 0.673 before to 0.761 on the event day, then fall to 0.700 after. These are descriptive model outputs; frame accuracy has not been established.

01 Question & workflow

Do rhetorical frames differ around conflict dates? Political posts can mix threats, success claims, and negotiations. Comparing those frames may help readers examine how public messaging changes around selected events.

Task. Given an Iran-related post, assign six independent frame scores, then compare the day before, day of, and day after an event's recorded start date.

1. **Collect & align**
   Prepared public-post CSV + UCDP-derived event workbook. Match Eastern Time calendar dates to event start dates at −1, 0, +1 days.
2. **Prepare the text**
   Use saved `content_text`; deduplicate by post ID. Preserve case and punctuation. Keep one post / event-date pair.
3. **Represent & classify**
   BART tokenizer → contextual representations → `facebook/bart-large-mnli`. Six natural-language frame descriptions; multi-label NLI, truncation enabled.
4. **Aggregate & compare**
   Average post scores within each event-date/window, then average those date means. Include every available post; a date does not need posts in all three windows.

The classifier sees post text only. Event headlines, fatalities, and severity do not enter the model. A post may receive several high scores.

02 Results

Mean frame score by window

Figure 1 · 0–1 scale

All available posts · 20 / 21 / 21 contributing dates · 40 / 40 / 49 post-event pairs (before / event day / after)



Each panel uses the same 0–1 y-axis. Scores are entailment scores, not calibrated probabilities, frequencies, or percentages of posts. Lines connect window means; no uncertainty intervals are estimated.

All available posts are included. Threat scores average 0.548 before, 0.524 on the event day, and 0.485 after. The windows contain different sets of event dates, so these descriptive differences may partly reflect coverage.

Different frame prompts may elicit different score levels; cross-frame rankings do not establish true prevalence.

03 Quality & interpretation

Preliminary · human evaluation pending

What was checked. All 87 score IDs match the post corpus; all six score columns are finite and within 0–1. Post/event pairs are unique and relative days match saved Eastern Time dates.

What remains untested. No human labels, held-out precision/recall/F1, or evaluated baseline are supplied. Data-integrity checks do not measure rhetorical accuracy.

A failure mode to investigate

A post praising the military operation receives a high Ceasefire / Ending score even though the text does not explicitly say the fighting has ended.

Ceasefire / Ending **0.802**

Inspect source post ↗ · Qualitative inspection, not a gold label.

Limits & responsible use

Association only. These selected windows cannot establish causation or predict military action. Same-day dates do not prove that a post preceded an event within that day.

Overlapping windows reuse posts and create dependence. Missing posts are not zero rhetoric. Linked headlines and media-only posts can distort attribution; truncated text can lose context.

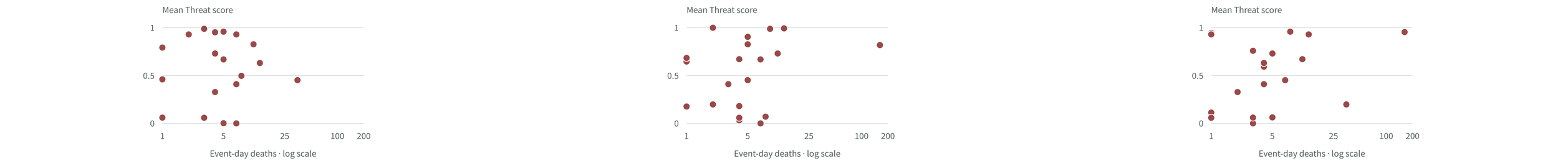
Practical implication. Use the scores to locate language worth inspecting alongside its source. Human validation and a non-event comparison are needed before stronger claims.

Event severity and rhetoric

Does stronger threatening language precede deadlier event dates? Compare Threat scores before, on, and after the event side by side. Each dot is one eligible event date; all panels share the same axes. The comparison menu also provides the four-frame view.

Overall mean Threat score: 0.548 before → 0.524 on the event day → 0.485 after. These all-data averages compare time windows. The scatterplots below ask a different question: whether higher Threat scores accompany higher event-day death counts.

Threat across all three windows · Day before: 19 dates · Event day: 20 dates · Day after: 20 dates. Each panel uses the dates with same-day fatality records and posts in that window.



Summed same-day best fatality estimate (log x-axis) versus mean model score (0–1). Horizontal whiskers show summed UCDP low–high estimates, not confidence intervals. Multi-day totals and windows with no posts are omitted. Spearman ρ describes rank association in this selected sample; it does not establish causation, prediction, or classifier accuracy.

Sources & reproducibility. Project repository, prepared CSVs & notebook · UCDP data access · BART-MNLI model · Matching public-post archive. The original post-archive snapshot and model revision remain unpinned; workbook release labels are recorded with the evidence. Counts refer to this repository, not the later portfolio dataset. UCDP requires attribution under CC BY 4.0; underlying post rights remain with their owners.

AI-use statement. The repository's notebook, prepared data, and saved scores preceded this poster task. OpenAI Codex inspected the rubric and code, built this page and reproducible aggregation, and checked joins, dates, counts, and rendered behavior. BART generated the saved analytical scores; it was not rerun for this poster. Sean directed the project. Human review and oral defense remain Sean's responsibility. See the repository for the fuller history.